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5 May 1983
SUPERSEDING
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3 July 1962

MILITARY SPECIFICATION
MOTORS, WINDSHIELD WIPER; PNEUMATIC, VACUUM, AND VALVES, CONTROL FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers air and vacuum windshield wiper motors and windshield wiper motor shut-off and speed control valves for use on military vehicles.

1.2 Classification. Windshield wiper motors and control valves shall be of the following types and classes, as specified (see 6.2):

Windshield wiper motors
Type I - vacuum operated
Type II - air pressure operated
  Class 1 - Pneumatic, universal
  Class 2 - Pneumatic

Speed and shut-off control valve
Type I - vacuum operated
Type II - air pressure operated

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2540

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2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS
MILITARY

MIL-P-116	- Preservation, Method of.
MIL-M-834	- Mittens Set, Extreme Cold Weather.

STANDARDS
FEDERAL

FED-STD-151	- Metals, Test Methods.
FED-STD-595	- Colors.

MILITARY

MIL-STD-12	- Abbreviations for Use on Drawings, Specifications, Standards and in Technical Documents.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MS 53017	- Motor, Windshield Wiper, Pneumatic, Universal.
MS 53018	- Valve, Pressure Regulating, Windshield Wiper Air Motor Control.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish the windshield wiper motors or control valves which shall be subjected to first article inspection (see 4.4 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent windshield wiper motors or control valves delivered to the Government shall conform to these samples in all their pertinent physical and performance attributes.

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3.2 Materials. All materials used in fabricating windshield wiper motors and control valves shall be new and shall conform to the requirements on applicable standards and drawings, and in this and other applicable specifications.

3.3 Design and construction. Design and construction shall be as specified herein, on MS 53017 and MS 53018, and other applicable drawings as specified by the acquisition activity (see 6.2). Type II, class 2 pneumatic wiper motors shall be physically and functionally interchangeable with type II, class 1 universal pneumatic wiper motors shown on MS 53017. Type I vacuum wiper motors shall conform to the design and construction on applicable drawings specified by the acquisition activity (see 6.2).

3.3.1 Manual operation. When specified on applicable drawings, windshield wiper motors shall be equipped with a handle or lever to permit manual operation. The design shall permit an operator wearing heavy gloves or mittens, such as "arctic" mittens defined in MIL-M-834, to operate the motor.

3.3.2 Adjustable cleaning arc. The sweep angle of type II, class 1 wiper motors shall be adjustable within a minimum range of 40° to 140°. Arc settings shall be made without disassembly of the wiper motor and shall be capable of being adjusted when mounted for intended operation. The arc setting shall remain fixed with in $\pm$ 3° after operation at 90 psi for 30 minutes as specified in 4.6.3.2.

3.3.3 Parking device. Type II, class 1 wiper motors shall incorporate, in combination with the shut-off and speed control device, provisions for moving the wiper blade to an extreme position when the control is turned to "off" position. Type II, class 2 and type I motors shall include such provisions when specified on applicable drawings, or by the procuring activity (see 6.2).

3.3.3.1 Manual parking override. Type II, class 1 motors shall have a right and left manual parking override of at least 30 degrees beyond the maximum cleaning arc.

3.3.4 Adjustable shaft. Type II, class 1 windshield wiper motors shall be designed to incorporate a reversible drive shaft, capable of being adjusted within a range from 1 to 4 inches, as shown on MS 53017. Reversal of the drive shaft shall permit either the large or small drive end of the shaft to be extended from the driving end of the wiper motor. Shaft adjustment or reversal shall be accomplished without disassembly of the wiper motor.

3.3.5 Shut-off and speed control. Type II pneumatic control valves shall be as specified on MS 53018. Type I vacuum control valves shall be as specified on applicable drawings specified by the acquisition activity (see 6.2).

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3.4 Performance.

3.4.1 Torque, speed and air consumption. Type II, classes 1 and 2 motors shall meet the requirements shown in table I. Type I motors shall meet the requirements shown in table II (see 4.6.2).

TABLE I. Ambient performance.

Air pressure, psi	Torque, lb-in. (min)	Speed, cpm (min)	Air consumption cfm (max)
20	12	85	0.35
30	20	95	0.50
50	40	100	0.70
80*	65*	105*	1.35*
90*	70*	105*	1.35*

* Applicable to type II, class 1 (universal) motor only

TABLE II. Ambient performance.

Vacuum, inches of mercury	Torque, lb-in (min)	Speed, cpm (min)
5	14	60
10	20	100
15	25	112
20	26	140
22	26	145

3.4.2 Speed control.

3.4.2.1 Pneumatic valves. Type II control valves shall regulate pneumatic wiper motor speed smoothly and uniformly over a speed range from 30 to 105 cycles per minute (see 4.6.3).

3.4.2.2 Vacuum valve. Type I control valves shall regulate vacuum wiper motor speed smoothly and uniformly over a speed range from 30 to 145 cycles per minute (see 4.6.3).

3.4.3 High temperature. When operated at ambient temperature of $160^{\circ} \pm 5^{\circ}F$, type II, classes 1 and 2 motors shall meet the requirements shown in table III. Type I motors in combination with type I control valves shall meet the requirements shown in table IV (see 4.6.4).

3.4.4 Low temperature. When operated at ambient temperature of $-20^{\circ} \pm 5^{\circ}F$, type II, classes 1 and 2 motors shall meet the requirements shown in table V. Type I motors shall be in combination with type I control valves shown in table VI (see 4.6.5).

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TABLE III. High temperature performance.

Air pressure, psi	Torque, lb-in. (min)	Speed, cpm (min)	Air consumption cfm (max)
20	12	90	0.30
30	20	100	0.45
50	40	105	0.65
80*	60*	110*	1.00*
90*	75*	110*	1.20*

* Application to type II, class 1 (universal) motor only

TABLE IV. High temperature performance.

Vacuum, inches of mercury	Torque, lb-in. (min)	Speed, cpm (min)
5	15	70
10	22	105
15	26	125
20	27	140
22	28	150

TABLE V. Low temperature performance.

Air pressure, psi	Torque, lb-in. (min)	Speed, cpm (min)	Air consumption, cfm (max)
20	12	40	0.55*
30	20	60	0.80*
50	37	75	1.30*
80*	60*	85*	1.75*
90*	68*	95*	1.85*

* Application to type II, class I (universal) motor only.

TABLE VI. Low temperature performance.

Vacuum, inches of mercury	Torque, lb-in. (min)	Speed, cpm (min)
10	5	10
15	7	15
20	10	20
22	12	26

3.4.5 Vibration resistance. Type II classes 1 and 2 motors shall meet the requirements of 3.4.1, table I; type I motors shall meet the requirements of 3.4.1, table II; type I and type II; type I and type II valves shall meet the requirements of 3.4.2 when subjected to the vibration criteria specified in method 201 of standard MIL-STD-202. Motors or valves shall show no evidence of breakage or loosening of components (see 4.6.6).

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3.4.6 Shock resistance. Type II, classes 1 and 2 motors shall meet the requirements of 3.4.1, table I; Type I motors shall meet the requirements of 3.4.2 when subjected to the shock criteria specified in method 202 of MIL-STD-202. Motors or valves shall show no evidence of breakage or loosening of components (see 4.6.7).

3.4.7 Endurance. Motors shall be capable of continuous operation for 300 hrs: Type II, classes 1 & 2 motors with an air pressure of 80 psi and a torque load of 50 inch/lbs; Type I motors with a vacuum equivalent of 15 inches of mercury and a torque load of 22 inch-lb. After 300 hours of operation, the motors will conform to the requirements specified in 3.4.1. Valves shall be capable of operating for 3 hours at 2000 cycles per hour at 80 psi. After three hours of operation the valves shall conform to the requirements specified in 3.4.2 (see 4.6.8).

3.5 Protective finish . Exterior surfaces of motors and valves, except pipe or screw threads, shall be painted or dyed olive drab. Color shall conform to color chip X24087 of FED-STD-595. Motors or valves shall be corrosion resistant and show no evidence of bare base material, undercoats, flaking paint or whitening when subjected to 200 hours of operation in accordance with method 101 of standard MIL-STD-202. After 200 hours operation in a corrosive environment, motors and valves shall conform to 3.4.1 & 3.4.2 (see 4.6.9).

3.5.1 Ferrous parts. The exposed metal surfaces of the motors and valves shall be corrosion resistant and show no evidence of corrosion when subjected to 200 hours exposure to method 101 of standard MIL-STD-202. After 200 hours exposure the motors and valves shall conform to the requirements of 3.4.1 and 3.4.2. (see 4.6.12).

3.5.2 Nonmetallic parts. All nonmetallic parts shall be so treated to resist the effects of salt spray. After treatment the non-metallic parts shall evidence no functional failure when subjected to 200 hours exposure to method 101 of MIL-STD-202 (see 4.6.9).

3.6 Marking. Marking shall consist of the part number shown on applicable drawing or Military Standard and manufacturer's identification in accordance with MIL-STD-130 (see 6.2).

3.7 Instructions. Illustrated instructions for installation, operation, setting and adjustment shall be packaged with each windshield wiper motor and control valve.

3.8 Definitions. Abbreviations are in accordance with MIL-STD-12.

3.9 Workmanship. Workmanship shall be of a quality which will assure a product free of burs, rust, scratches, chips, sharp edges, or other defects which will affect serviceability or appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use any facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
c. Acceptance test (see 4.5.2).
d. Control test (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all tests shall be conducted at an ambient still air temperature of $+77^{\circ} \pm 5^{\circ}$F. Unless otherwise specified, a $90^{\circ} \pm 5^{\circ}$ cleaning arc shall be used in testing type II, class 1 motors.

4.3.1 Calibration of test equipment. Unless otherwise specified, test equipment shall be accurate to within 10 percent of the tolerance on the characteristic to be measured. Calibration of test equipment shall be conducted at intervals which serve to establish required accuracy. Records of calibration shall be made available to the Government. The Government inspector shall refuse to allow inspection where accuracy of test equipment has not been established to the satisfaction of the Government. Calibration of inspection and test equipment shall be in accordance with MIL-STD-45662.

4.4 First article inspection. First article inspections are categorized as preproduction inspection or initial production inspection. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply windshield wipers or control valves that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two sample windshield wipers or control valves. Preproduction inspection shall consist of examination as specified in 4.5.2 and tests specified in table VII.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select 8 assemblies, from the first 50 motors or valves produced under the production contract (see 6.2) for initial production inspection. Initial production units shall be examined as specified in 4.5.2 and tested as specified in table VII.

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TABLE VII. Classification of tests.

Title	Requirement	Preproduction initial prod.	Acceptance tests 100%	Control tests
Pretest break-in		4.6.1	4.6.1	
Stall torque, speed and air consumption	3.4.1	4.6.2	4.6.2	
Speed control test	3.3.5	4.6.3	4.6.3	
Parking test	3.3.3	4.6.3.1		
Adjustable cleaning arc	3.3.2	4.6.3.2		4.6.3.2
High temperature	3.4.3	4.6.4		
Low temperature	3.4.4	4.6.5		
Vibration	3.4.5	4.6.6		
Shock	3.4.6	4.6.7		4.6.7
Endurance	3.4.7	4.6.8	4.6.8	
Corrosion test	3.5	4.6.9	4.6.9	4.6.9

4.4.3 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval until corrective action by the contractor has been approved by the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all motors or valves of one type and class, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.5.2 Acceptance tests.

4.5.2.1 Sampling for acceptance testing. Each assembly from each lot (see 4.5.1.1) shall be subjected to all acceptance tests as specified in table VII.

4.5.2.2 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined for conformance to the following acceptable quality levels (AQL's) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.3 Classification of defects. For examination purposes, defects shall be classifeid as specified in table VIII.

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TABLE VIII. Classification of defects.

Categories	Defects	Method of inspection
Critical	None defined	
Major	AQL 1.0	
101	Dimensions affecting interchangeability (see 3.3).	Gage
102	Nonconformance in design and construction (see 3.3).	Manual & Gage
Minor	AQL 2.5	
201	Dimension not affecting interchangeability (see 3.3).	Gage
202	Improper marking (see 3.6).	Visual
203	Faulty workmanship (see 3.9).	Visual

4.5.2.4 Failure. Any assembly that fails to pass any acceptance test, or exhibits defects in excess of AQL specified, shall be rejected. Rejected units may be reworked or repaired and resubmitted for acceptance testing.

4.5.3 Control tests.

4.5.3.1 Control inspection. When specified (see 6.2), control test samples shall be selected at the rate of 5 of each 100 motors or valves produced. Motors or valves selected shall be examined for the defects specified in 4.5.2.2, and subsequently tested as specified in 4.6.1, 4.6.2, and 4.6.3.

4.5.3.2 Failure. Failure of a control test sample to pass any specified test shall be cause for the the Government to refuse to accept subsequent lots until it has been proven, to the satisfaction of the Government, that the faults revealed by the tests have been corrected.

4.6 Methods of inspection.

4.6.1 Pretest break-in. Prior to test, the motors shall be operated as follows: Type II, class 1 motors shall be operated continuously for 3 hours with an air pressure of 80 ± 5 psi and type II; class 2 motors shall be operated continuously for 3 hours with an air pressure of 45 ± 5 psi. Type I motors shall be operated continuously for 3 hours at vacuum equivalent of 22 inches of mercury. Break-in operation shall be conducted with a suitable wiper arm and blade assembled to the motor. The wiper blade shall be in contact with glass which shall be sprayed wet with water during the break-in period.

4.6.2 Stall torque, speed and air consumption test. To determine conformance to 3.4.1, the motors shall be operated at the applicable air pressure or vacuum specified in table I or table II. The stall torque and speed, and the air consumption only for type II, classes 1 and 2 motors shall be recorded.

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4.6.3 Speed control test. Type II, class 1 motors shall be tested with an air pressure of 90 psi, type II, class 2 motors shall be tested with an air pressure of 50 psi, and type I motors shall be tested with a vacuum equivalent of 22 inches of mercury. For each motor tested, the average of five trials shall be recorded for the minimum speed. The motor shall be observed for smooth and gradually increasing speed of operation.

4.6.3.1 Parking test. Parking and override operation shall be checked, as applicable, to determine conformance to 3.3.3 and 3.3.3.1 at each pressure or vacuum setting specified in tables I and II.

4.6.3.2 Adjustable cleaning arc test. To determine conformance to 3.3.2, cleaning arc of type II, class 1 wiper motors shall be set at 40, 100, and 140 degrees, and operated at 90 psi for 30 minutes at each arc setting.

4.6.4 High temperature test. To determine conformance to 3.4.3, specimens (and associated control valves for type II, class 1 motor test) shall be conditioned in a heat cabinet for 5 hours at 160° ± 5°F. While at this temperature, the torque, speed and air consumption for type II, classes 1 and 2 motors shall be recorded.

4.6.5 Low temperature test. To determine conformance to 3.4.4, specimens (and associated control valves for type II, class 1 motor test) shall be conditioned in a cold cabinet for 5 hours at minus 20° ± 5°F. While at this temperature, the torque, speed, and air consumption for type II, class 1 motors shall be recorded. The torque and speed for type II, class 2 and type I motors shall be recorded.

4.6.6 Vibration test. To determine conformance to 3.4.5, motors or valves shall be mounted by the means intended for installation, and rigidly mounted to the vibrating machine. The test method shall conform to method 201 of MIL-STD-202. Upon completion, the motors or valves shall then be tested for conformance to 3.4.1 and 3.4.2.

4.6.7 Shock test. To determine conformance to 3.4.6, motors or valves shall be mounted by the means intended for installation. The test method shall conform to method 202 of MIL-STD-202. Test specimens shall be dropped 3 times in each plane of the 3 major axes. The acceleration shall be 50 gravity units in each of the three major axes. Upon completion, the motors or valves shall then be tested for conformance to 3.4.1 and 3.4.2.

4.6.8 Endurance test. To determine conformance to 3.4.7, air pressure of 80 psi and a torque load of 50 inch-lb shall be imposed on type II, class 1 motors, and air pressure of 45 psi and a torque of 30 inch-lbs shall be imposed on type II, class 2 motors.

4.6.9 Corrosion resistance test. To determine conformance to 3.5, 3.5.1, and 3.5.2, motors and valves shall be tested in accordance with method 101 of MIL-STD-202 for 200 hours. Upon completion, the motors and valves shall be tested for conformance to 3.4.1 and 3.4.2.

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The windshield wiper motors and control valves covered by this specification are intended for use in military vehicles, construction equipment, industrial trucks and tractors, marine application and other similar uses. Services and repair parts for motors and valves covered herein are not intended to be carried in the supply system.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type and class (see 1.2).
c. MS part numbers or applicable drawings (see 3.3).
d. Any special lot size (see 4.5.1).
e. Selection of applicable levels of preservation, packaging, and packing (see 5.1).
f. Whether control tests be specified (see 4.5.4).
g. First article if required (see 3.1).

Custodian:
Army - AT
Navy - YD

Preparing activity:
Army - AT

Project No. 2540-0323

User activity:
Navy - MC
DLA - CS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-12185C	Motor, Windshield Wiper; Pneumatic, Vacuum and Valves, Control For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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